多文档摘要
Multi-Document Summarization

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1.概述

- **任务定义：**从包含多份文档的文档集合中生成能够概括这些文档中心内容的摘要。

- **评估方法：**
  - 自动评估：Rouge指标（Rouge-1，Rouge-2与Rouge-L）
  - 人工评估：根据摘要的冗余度、信息量等进行评分

2.难点

多文档摘要相比于单文档摘要的难点

- **数据噪声大：**对于一个文档集合来说，文档越多，其包含的主题、噪声也越多，因此提取摘要的难度也越大。
- **模型设计难度提升：**多文档摘要的输入众多，可能包含上百篇文章输入，因此直接使用端到端的模型去训练并不现实。
- **数据稀缺：**相比于单文档，多文档摘要的数据集十分稀缺，在18年之前只有DUC/TAC发布的多文档摘要数据集，每期也只有百余篇。直到2018年才提出了第一个大规模的多文档摘要数据集WikiSum。

3.数据集

数据集	规模	特点
DUC-2001~2007 TAC-2008~2015	百余篇	数据量非常小
WikiSum（2018）	1M+实例	每个实例平均包含525个段落
Multi-News（2019）	56K+实例	每个实例包含2-10篇文章与对应摘要
GameWikiSum（2020）	14K+实例	标准摘要较长，每个实例包含2-84篇文章

4.现有方法总结

- MDS
  - **抽取式多文档摘要**
    - 有监督抽取
      - 基于RNNs/Transformer [1]
      - 基于GCN [2,3]
    - 无监督抽取
      - 基于句子排序
      - 基于稀疏重建 [5]
  - **生成式多文档摘要**
    - 小规模数据集时代
      - 基于短语融合 [4]
      - 迁移学习 [6,7]
    - 大规模数据集时代
      - 未考虑文档之间的关系 [8,9]
      - 考虑文档之间的关系 [10]
  - **生成&抽取**
    - 先抽取再生成
    - 抽取生成联合训练让模型更好习得不同粒度的表示 [11]

4. 现有方法总结

- [1]2015 AAAI: Ranking with Recursive Neural Networks and Its Application to Multi-Document Summarization
- [2]2017 CoNLL：Graph-based Neural Multi-Document Summarization
- [3]2020 ACL：Heterogeneous Graph Neural Networks for Extractive Document Summarization
- [4]2015 ACL：Abstractive Multi-Document Summarization via Phrase Selection and Merging
- [5]2016 COLING：An Unsupervised Multi-Document Summarization Framework Based on Neural Document Model
- [6]2018 arXiv：Towards a Neural Network Approach to Abstractive Multi-Document Summarization
- [7]2018 EMNLP：Adapting the Neural Encoder-Decoder Framework from Single to Multi-Document Summarization
- [8]2018 ICLR：Generating Wikipedia by Summarizing Long Sequences
- [9]2019 ACL：Multi-News: A Large-Scale Multi-Document Summarization Dataset and Abstractive Hierarchical Model
- [10]2019 ACL：Hierarchical Transformers for Multi-Document Summarization
- [11]2020 ACL：Multi-Granularity Interaction Network for Extractive and Abstractive Multi-Document Summarization

5.最新论文分享

Heterogeneous Graph Neural Networks for Extractive Document Summarization

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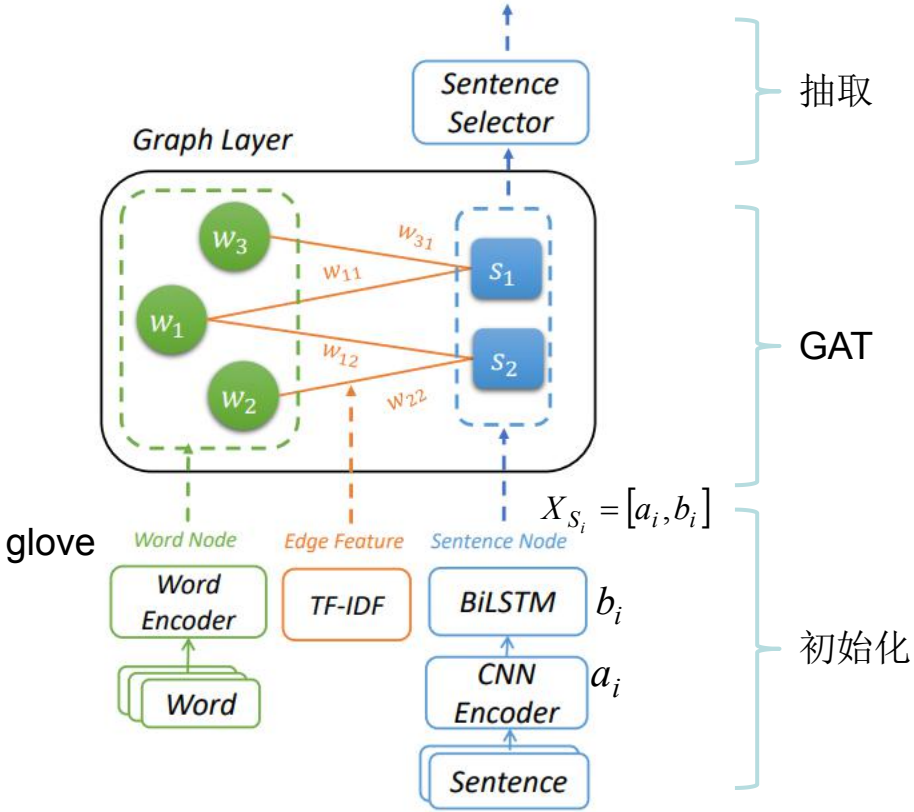
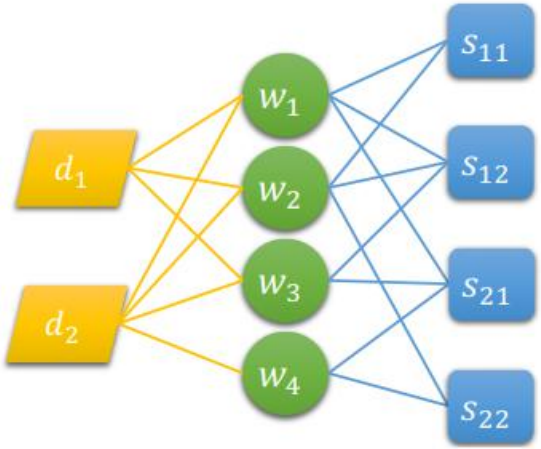
2020 ACL

- 动机：利用异质图神经网络来做抽取式文本摘要。





Model	R-1	R-2	R-L
First-1	25.44	7.06	22.12
First-2	35.70	10.28	31.71
First-3	40.21	12.13	37.13
ORACLE	52.32	22.23	47.93
LexRank* (Erkan and Radev, 2004)	41.77	13.81	37.87
TextRank* (Mihalcea and Tarau, 2004)	41.95	13.86	38.07
MMR* (Carbonell and Goldstein, 1998)	44.72	14.92	40.77
PG† (Lebanoff et al., 2018)	44.55	15.54	40.75
BottomUp† (Gehrmann et al., 2018)	45.27	15.32	41.38
Hi-MAP† (Fabbri et al., 2019)	45.21	16.29	41.39
HSG	45.66	16.22	41.80
HSG + Tri-Blocking	44.92	15.59	40.89
HDSG	**46.05**	**16.35**	**42.08**
HDSG + Tri-Blocking	45.55	15.78	41.29

Table 4: Results on the test set of Multi-News.

Hierarchical Transformers for Multi-Document Summarization

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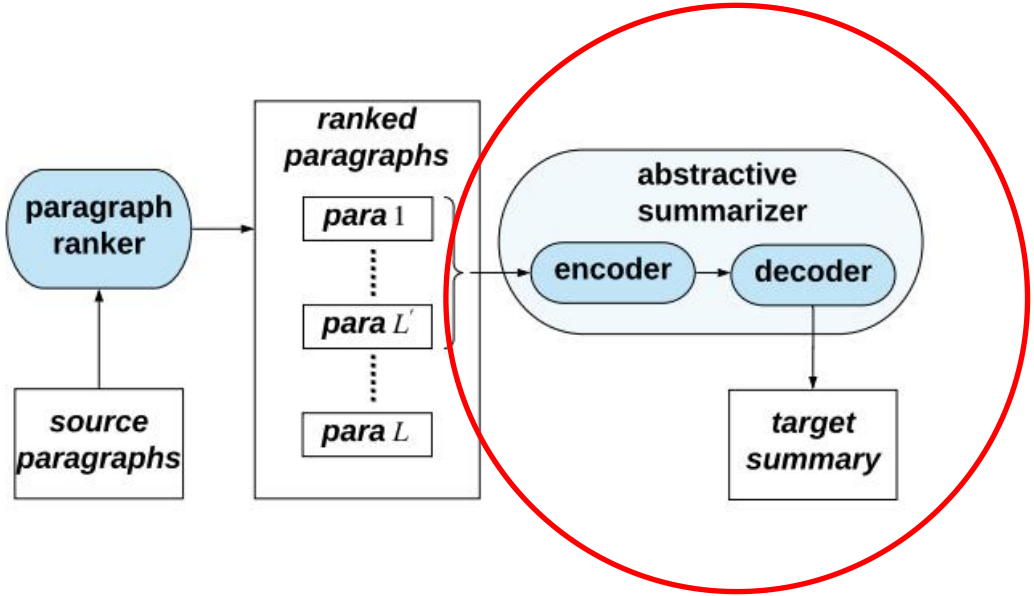
2019 ACL

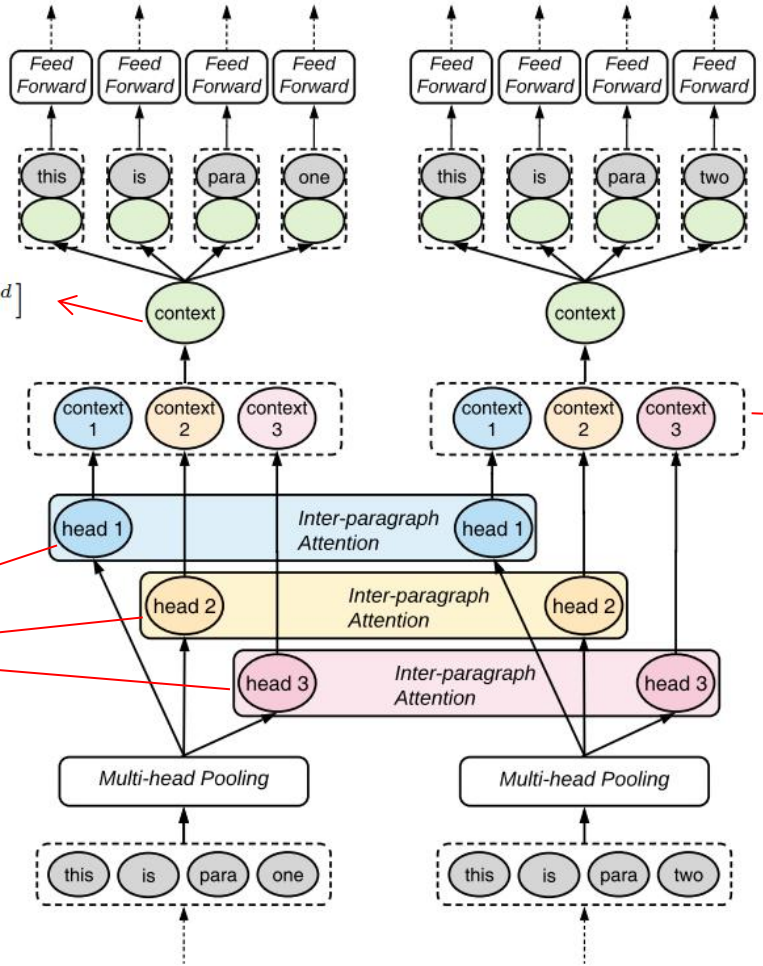
动机：在大规模多文档摘要数据集上探索如何**利用多文档间的关系**来助力摘要的生成。



$$g_{ij} = W_{o2}\mathrm{ReLU}(W_{o1}(x_{ij}^{l-1} + c_i))$$
$$x_{ij}^{l} = \mathrm{LayerNorm}(g_{ij} + x_{ij}^{l-1})$$

$$c_i = W_c[context_i^1; \cdots ; context_i^{n_{head}}]$$

$$a_{ij}^z = W_a^z x_{ij}^{l-1}$$
$$b_{ij}^z = W_b^z x_{ij}^{l-1}$$
$$\hat{a}_{ij}^z = exp(a_{ij}^z)/\sum^{n} exp(a_{ij}^z)$$
$$head_i^z = \mathrm{LayerNorm}(W_c^z \sum_{j=1}^{n} a_{ij}^z b_{ij}^z)$$



$$q_i^z = W_q^z head_i^z$$
$$k_i^z = W_k^z head_i^z$$
$$v_i^z = W_v^z head_i^z$$
$$context_i^z = \sum_{i=1}^{m} \frac{exp(q_i^{z^T} k_{i'}^z)}{\sum_{o=1}^{m} exp(q_i^{z^T} k_o^z)} v_{i'}^z$$

Model	ROUGE-1	ROUGE-2	ROUGE-L
Lead	38.22	16.85	26.89
LexRank	36.12	11.67	22.52
FT (600 tokens, no ranking)	35.46	20.26	30.65
FT (600 tokens)	40.46	25.26	34.65
FT (800 tokens)	40.56	25.35	34.73
FT (1,200 tokens)	39.55	24.63	33.99
T-DMCA (3000 tokens)	40.77	25.60	34.90
HT (1,600 tokens)	**40.82**	**25.99**	35.08
HT (1,600 tokens) + Similarity Graph	40.80	25.95	35.08
HT (1,600 tokens) + Discourse Graph	40.81	25.95	**35.24**
HT (train on 1,600 tokens/test on 3000 tokens)	**41.53**	**26.52**	**35.76**

Multi-Granularity Interaction Network for Extractive and Abstractive Multi-Document Summarization

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2020 ACL

动机：抽取&生成联合训练让模型更好习得不同粒度的表示。

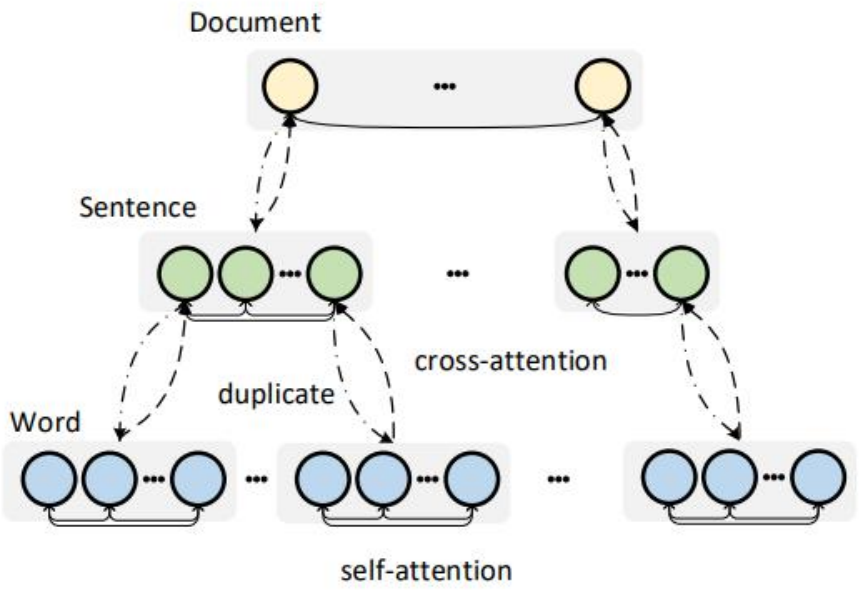


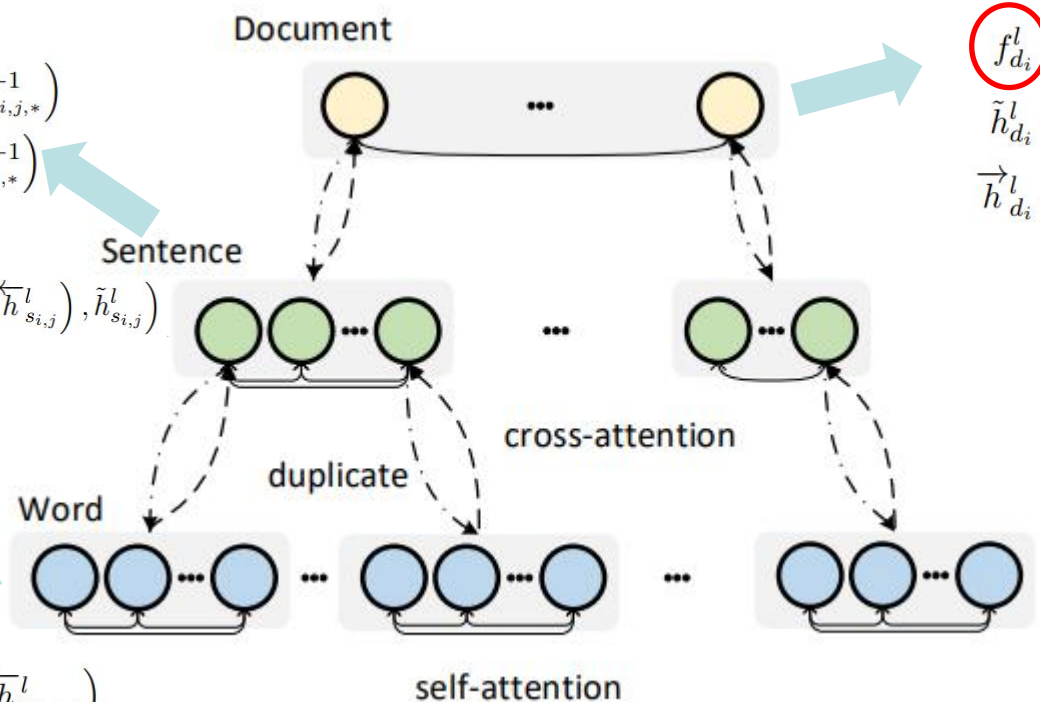
Figure 1: The overview of the multi-granularity encoder layer.

$$\overrightarrow{h}^{l}_{s_{i,j}} = \mathrm{MHAtt}\left(h^{l-1}_{s_{i,j}}, h^{l-1}_{w_{i,j,*}}\right)$$

$$\tilde{h}^{l}_{s_{i,j}} = \mathrm{MHAtt}\left(h^{l-1}_{s_{i,j}}, h^{l-1}_{s_{i,*}}\right)$$

$$\overleftarrow{h}^{l}_{s_{i,j}} = h^{l-1}_{d_i}$$

$$f^{l}_{s_{i,j}} = \mathrm{Fusion}\left(\mathrm{Fusion}\left(\overrightarrow{h}^{l}_{s_{i,j}}, \overleftarrow{h}^{l}_{s_{i,j}}\right), \tilde{h}^{l}_{s_{i,j}}\right)$$



$$f^{l}_{d_i} = \mathrm{Fusion}\left(\tilde{h}^{l}_{d_i}, \overrightarrow{h}^{l}_{d_i}\right)$$

$$\tilde{h}^{l}_{d_i} = \mathrm{MHAtt}\left(h^{l-1}_{d_i}, h^{l-1}_{d_*}\right)$$

$$\overrightarrow{h}^{l}_{d_i} = \mathrm{MHAtt}\left(h^{l-1}_{d_i}, h^{l-1}_{s_{i,*}}\right)$$

$$f^{l}_{w_{i,j,k}} = \mathrm{Fusion}\left(\tilde{h}^{l}_{w_{i,j,k}}, \overleftarrow{h}^{l}_{w_{i,j,k}}\right)$$

$$\tilde{h}^{l}_{w_{i,j,k}} = \mathrm{MHAtt}\left(h^{l-1}_{w_{i,j,k}}, h^{l-1}_{w_{i,j,*}}\right)$$

$$\overleftarrow{h}^{l}_{w_{i,j,k}} = h^{l-1}_{s_{i,j}}$$

$$z = \sigma\left([x; y] W_f + b_f\right)$$

$$\mathrm{Fusion}(x, y) = z\, x + (1 - z)\, y$$

$$\tilde{h} = \mathrm{LayerNorm}\left(h^{l-1} + f^{l}\right)$$

$$h^{l} = \mathrm{LayerNorm}(\tilde{h} + \mathrm{FFN}(\tilde{h}))$$

抽取式摘要：

$$\tilde{y}_s = \sigma\left(o_s W_o + b_o\right)$$

生成式摘要：

$$\tilde{g} = \text{LayerNorm}\left(g^{l-1} + \text{MSAttn}\left(g^{l-1}, g^{l-1}\right)\right)$$

$$c = \text{LayerNorm}\left(\tilde{g} + \text{MSAtt}\left(\tilde{g}, o_w\right)\right)$$

$$g^l = \text{LayerNorm}\left(c + \text{FFN}(c)\right)$$

$$p_t^g = \text{softmax}\left(g_t^{L_2} W_g + b_g\right)$$

$$\varepsilon_t = \text{softmax}(g_t^{L_2} o_w^\top + b_\varepsilon)$$

$$p_t^c = \sum \varepsilon_t z_{i,j,k}^\top$$

$$\eta_t = \sigma\left(g_t^{L_2} W_\eta + b_\eta\right)$$

$$p_t = \eta_t * p_t^g + (1 - \eta_t) * p_t^c$$

Loss：

$$L_{mix} = L_{abs} + \lambda L_{ext}$$

Model	R-1	R-2	R-SU4
Lead-3	39.41	11.77	14.51
LexRank (Erkan and Radev, 2004)	38.27	12.70	13.20
TextRank (Mihalcea and Tarau, 2004)	38.44	13.10	13.50
MMR(Carbonell and Goldstein, 1998)	38.77	11.98	12.91
HIBERT (Zhang et al., 2019)	43.86	14.62	18.34
PGN (See et al., 2017)	41.85	12.91	16.46
CopyTransformer(Gehrmann et al., 2018)	43.57	14.03	17.37
Hi-MAP(Fabbri et al., 2019)	43.47	14.89	17.41
HF(Liu and Lapata, 2019)	43.85	15.60	18.80
MGSum-*ext*	44.75	15.75	19.30
MGSum-*abs*	**46.00**	**16.81**	**20.09**
oracle ext	49.02	29.78	29.19

Table 2: ROUGE F1 evaluation results on the Multi-News test set.

感谢聆听